Linear Optimization (311.160)
Exercise sheet 1

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Exercise 1.1: definitions and comparison

The AI answers below use the maximization convention from the current lecture notes. For every item, the comparison refers to *Lecture 2 – 6.10.2026*, PDF pages 5, 8 and 9. Each answer is followed by the corresponding lecture definition and a check.

1. **Standard form.**

   AI answer: A linear program in standard form is

   $$\max\ c^{\mathsf{T}}x \quad \text{subject to} \quad Ax = b, \quad x \geq 0,$$

   where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$ and $c, x \in \mathbb{R}^n$. The objective is linear, the constraints $Ax = b$ are equalities, and all variables are nonnegative.

   Lecture comparison (p. 5): The notes give precisely this maximization problem as standard form. ✓ Match. Other textbooks may use a different convention, but that is not the convention used here.

2. **Feasible or infeasible vector.**

   AI answer: A vector $x \in \mathbb{R}^n$ is feasible if it satisfies every constraint, that is, $Ax = b$ and $x \geq 0$. It is infeasible if at least one constraint fails.

   Lecture comparison (p. 8): The notes call a vector satisfying all constraints feasible and any other vector infeasible. ✓ Match.

3. **Feasible set.**

   AI answer: The feasible set contains all feasible vectors:

   $$C = \{x \in \mathbb{R}^n : Ax = b,\ x \geq 0\}.$$

   Lecture comparison (p. 8): The notes define the same set C. ✓ Match.

4. **Feasible or infeasible linear program.**

   AI answer: The LP is feasible if $C \neq \varnothing$, meaning that some vector satisfies all constraints. It is infeasible if $C = \varnothing$.

   Lecture comparison (p. 8): The distinction is exactly $C \neq \varnothing$ versus $C = \varnothing$. ✓ Match.

5. **Optimal value.**

   AI answer: For a maximization LP, the optimal value is

   $$\gamma = \sup\{c^\mathsf{T} x : x \in C\} \in \mathbb{R} \cup \{-\infty, +\infty\}, \qquad \sup \varnothing = -\infty.$$

   Thus an infeasible LP has value $-\infty$, a feasible unbounded LP has value $+\infty$, and a feasible bounded LP has a finite real optimal value.

   Lecture comparison (pp. 8–9): The notes use this supremum and the same empty-set convention, then list these three cases. ✓ Match. The supremum definition does not itself assert that a maximizer exists.

6. **Bounded or unbounded linear program.**

   AI answer: A *feasible* maximization LP is unbounded if, for every $M \in \mathbb{R}$, there is an $x \in C$ with $c^\mathsf{T} x > M$. It is bounded if some $M \in \mathbb{R}$ satisfies $c^\mathsf{T} x \leq M$ for every $x \in C$.

   Lecture comparison (p. 8): These are exactly the two conditions in the notes. ✓ Match. Both definitions assume feasibility; an infeasible LP is a separate case. Boundedness here concerns the objective, not whether the feasible set is a bounded set.

7. **Optimal solution.**

   AI answer: A vector $x^* \in C$ is an optimal solution, or maximizer, if

   $$c^\mathsf{T} x^* \geq c^\mathsf{T} x \quad \text{for every } x \in C.$$

   Equivalently, it is feasible and attains the optimal value.

   Lecture comparison (p. 9): The notes use the same comparison with every feasible vector and also call such a vector a maximizer. ✓ Match.

Exercise 1.2: choosing information sources

Let $x_i \in \{0, 1\}$ indicate whether option i is chosen:

i	Option	Value	Time (min)
1	Ask a colleague	4	5
2	Attend the first lecture	8	9
3	Read the Moodle course description	7	7
4	Read the LV-Karte	16	15
5	Do nothing	40	30

Each option is chosen at most once. Values and times add across chosen options.

Main interpretation: at least two actual information sources. Doing nothing provides no information and therefore does not count as a source. The requirement to use at least two sources is mandatory, not waived by option 5. The binary optimization model is

$$\begin{aligned} \max \quad & 4x_1 + 8x_2 + 7x_3 + 16x_4 + 40x_5 \\ \text{subject to} \quad & 5x_1 + 9x_2 + 7x_3 + 15x_4 + 30x_5 \leq 30, \\ & x_1 + x_2 + x_3 + x_4 \geq 2, \\ & x_i \in \{0, 1\}, \qquad i = 1, \ldots, 5. \end{aligned}$$

Option 5 already consumes the entire time budget. Every actual source takes positive time, so $x_5 = 1$ contradicts the requirement to use two sources. Hence every feasible choice has $x_5 = 0$; retaining it in the model makes that consequence explicit. If all five listed options are instead counted toward the condition $\sum_{i=1}^{5} x_i \geq 2$, the feasible choices are unchanged because option 5 still cannot be combined with another option.

As a check, the best feasible choice is $(x_1, x_2, x_3, x_4, x_5) = (1, 1, 0, 1, 0)$: it uses three information sources, takes $5 + 9 + 15 = 29$ minutes, and has value $4 + 8 + 16 = 28$.

Alternative interpretation: doing nothing is an opt-out. The inclusion of option 5 could mean that the student may decline to gather information altogether. That reading needs an explicit exception to the two-source requirement. Under that assumption, replace the source constraint by

$$x_1 + x_2 + x_3 + x_4 \geq 2(1 - x_5), \qquad x_i + x_5 \leq 1 \quad (i = 1, \ldots, 4),$$

while keeping the objective, time budget and binary domains. Then either at least two sources are used, or only option 5 is chosen. The latter gives value 40 in 30 minutes and is optimal for this alternative model. It does *not* satisfy the sheet's literal two-source requirement, so the main model above is my formulation without an opt-out exception.

This is a binary integer linear optimization problem, following the selection model in Lecture 1, p. 8. No conversion to standard form is needed.

Exercise 1.3: FreshShine

Define the monthly production quantities

$$\begin{aligned} x_1 &= \text{units of Citrus Shower Gel,} \\ x_2 &= \text{units of Citrus Shampoo,} \\ x_3 &= \text{units of Citrus Body Lotion.} \end{aligned}$$

For the LP formulation, quantities are divisible and $x_1, x_2, x_3 \in \mathbb{R}_{\geq 0}$. If whole units were required, integer domains would be an additional assumption.

Since all produced units can be sold, the profit in euros is $3x_1 + 5x_2 + 4x_3$. The model is

$$\begin{aligned} \max \quad & 3x_1 + 5x_2 + 4x_3 & \\ \text{subject to} \quad & x_1 + 2x_2 + x_3 \leq 170 & \text{(machine A, hours/month),} \\ & x_1 + x_2 + 2x_3 \leq 150 & \text{(machine B, hours/month),} \\ & 3x_2 + x_3 \leq 180 & \text{(machine C, hours/month),} \\ & x_1, x_2, x_3 \geq 0. & \end{aligned}$$

Citrus Shower Gel uses no machine C time, which explains its zero coefficient in that constraint. There is no sales constraint because the sheet states that every produced unit can be sold. This follows the production-model structure in Lecture 1, pp. 3–4: nonnegative quantities, total profit, and one constraint for each limited resource. No conversion to standard form is needed.